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Testing VPC Connectivity



yakubu abdullahi yusuf

```
[ec2-user@ip-10-0-0-187 ~]$ curl https://learn.nextwork.org/projects/aws-host-a-website-on-s3
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />

    <title>NextWork - Host a Website on Amazon S3</title>
    <meta content="Let's host your very own website on Amazon S3!" name="description" />
    <meta
      content="NextWork - Host a Website on Amazon S3"
      property="og:title"
    />
    <meta
      content="Let's host your very own website on Amazon S3!"
      property="og:description"
    />
    <meta
      content="NextWork - Host a Website on Amazon S3"
      property="twitter:title"
    />
```



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Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a service that lets you create your own private network inside AWS where you can run cloud resources like EC2 instances. It is useful because it gives you full control over networking and security.

How I used Amazon VPC in this project

In today's project, I used Amazon VPC to configure inbound and outbound rules to ensure reliable connectivity between my instances and my instance to the internet.

One thing I didn't expect in this project was...

One thing I didn't expect in this project was how many different networking and security components such as subnets, route tables, security groups, network ACLs, and gateways have to work together just to enable basic connectivity.

This project took me...

This project took me thirty minutes to complete.



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Connecting to an EC2 Instance

Connectivity is how well different parts of a network talk to each other and with external networks. It's very important because connectivity is also about how data flows smoothly across your network, powering everything from simple web hosting on the Internet to complex operations.

My first connectivity test was whether I could connect to my public instance using AWC Console connect.

```
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
c2-user@ip-10-0-0-187 ~]$
```

i-0af932dc6787a8986 (NextWork Public Server)

PublicIP: 18.188.41.124 PrivateIP: 10.0.0.187



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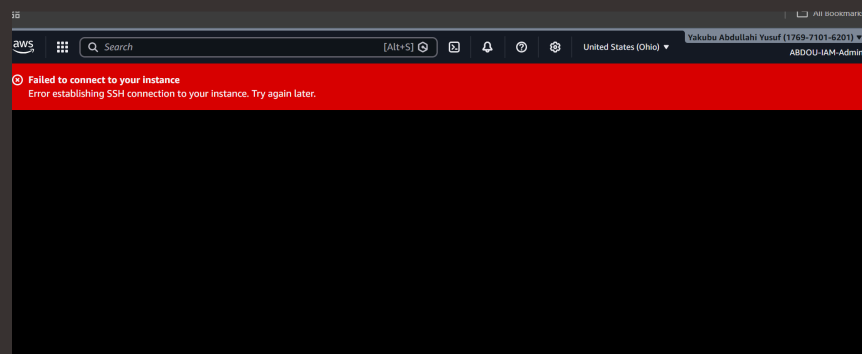
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EC2 Instance Connect

I connected to my EC2 instance using EC2 Instance Connect, which is an AWS feature that allows a secure connection to an EC2 instance using SSH directly from the AWS Console or a CLI without permanently storing SSH keys on the instance.

My first attempt at getting direct access to my public server resulted in an error, because of a rule that is missing in the security group of my public instance.

Are you wondering what rule was missing, I fixed this error by adding an SSH inbound rule with a port 22.





Connectivity Between Servers

Ping is a tool that helps verify connection between two computers or servers. I used ping to test the connectivity between my public EC2 instance and private EC2 instance.

The ping command I ran was ping 10.0.1.138

The first ping returned PING 10.0.1.138 (10.0.1.138) 56(84) bytes of data. This meant that the instance could see and reach the private EC2 instance on the network and attempting to send packets to it but couldn't due to an error.



```
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
2-user@ip-10-0-0-187 ~]$ ping 10.0.1.138
P 10.0.1.138 (10.0.1.138) 56(84) bytes of data.
```



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Troubleshooting Connectivity

I troubleshooted this by adding new inbound and outbound rule to my private instance's security group.

```
ec2-user@ip-10-0-0-187 ~]$ ping 10.0.1.138
PING 10.0.1.138 (10.0.1.138) 56(84) bytes of data.
64 bytes from 10.0.1.138: icmp_seq=889 ttl=127 time=0.748 ms
64 bytes from 10.0.1.138: icmp_seq=890 ttl=127 time=0.754 ms
64 bytes from 10.0.1.138: icmp_seq=891 ttl=127 time=0.740 ms
64 bytes from 10.0.1.138: icmp_seq=892 ttl=127 time=0.739 ms
64 bytes from 10.0.1.138: icmp_seq=893 ttl=127 time=0.733 ms
64 bytes from 10.0.1.138: icmp_seq=894 ttl=127 time=0.747 ms
64 bytes from 10.0.1.138: icmp_seq=895 ttl=127 time=0.760 ms
64 bytes from 10.0.1.138: icmp_seq=896 ttl=127 time=0.751 ms
64 bytes from 10.0.1.138: icmp_seq=897 ttl=127 time=0.740 ms
64 bytes from 10.0.1.138: icmp_seq=898 ttl=127 time=0.744 ms
64 bytes from 10.0.1.138: icmp_seq=899 ttl=127 time=0.742 ms
64 bytes from 10.0.1.138: icmp_seq=900 ttl=127 time=0.761 ms
64 bytes from 10.0.1.138: icmp_seq=901 ttl=127 time=0.814 ms
64 bytes from 10.0.1.138: icmp_seq=902 ttl=127 time=0.726 ms
64 bytes from 10.0.1.138: icmp_seq=903 ttl=127 time=0.726 ms
64 bytes from 10.0.1.138: icmp_seq=904 ttl=127 time=0.734 ms
64 bytes from 10.0.1.138: icmp_seq=905 ttl=127 time=0.745 ms
```



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Connectivity to the Internet

Curl is used to transfer data to or from a server.

I used curl to test the connectivity between my instance and websites.

Ping vs Curl

Ping and curl are different because Ping asks if a server is reachable and Curl asks if it can get data from a server.



Connectivity to the Internet

I ran the curl command `curl https://learn.nextwork.org/projects/aws-host-a-website-on-s3` which returned the HTML content of the webpage, confirming that my instance had successful internet access and could retrieve data from a web server.

```
[ec2-user@ip-10-0-0-187 ~]$ curl https://learn.nextwork.org/projects/aws-host-a-website-on-s3
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